Actuarial Exam P Study Guide — Phase 1: General Probability (Days 1–10)

Welcome to Phase 1 of your **Actuarial Exam P** preparation. Phase 1 covers **General Probability**, accounting for **23–30%** of the questions on the official Society of Actuaries (SOA) Exam P.

This comprehensive guide is structured day-by-day according to your study schedule. Each day features:

1. **Core Objectives & Theoretical Lesson**
2. **Worked Examples with Solutions**
3. **10 Practice Problems**

All answers and step-by-step solutions for the 100 practice problems across Days 1–10 are collected at the end of this document in the **Complete Answer Key & Solutions** section.

Day 1: Sample Spaces, Set Operations, & Event Algebra (Session 1A.1)

1. Core Objectives & Theoretical Lesson

- **Sample Space (S or Ω):** The set of all possible outcomes of a random experiment.
- **Outcome (ω):** A single element of the sample space ($\omega \in S$).
- **Event (A):** Any subset of the sample space ($A \subseteq S$). An event A occurs if the outcome $\omega \in A$.
- **Null Event ($\varnothing$):** The empty set containing no outcomes.

Fundamental Set Operations

Let A and B be events in a sample space S:

1. **Union ($A \cup B$):** Outcomes in A, or B, or both (at least one occurs).
2. **Intersection ($A \cap B$):** Outcomes in both A and B simultaneously.
3. **Complement (A^c or A'):** Outcomes in S that are not in A ($A^c = S \setminus A$).
4. **Set Difference ($A \setminus B = A \cap B^c$):** Outcomes in A but not in B.
5. **Symmetric Difference ($A \Delta B$):** Outcomes in A or B, but not both:

$$A \Delta B = (A \cap B^c) \cup (B \cap A^c) = (A \cup B) \setminus (A \cap B)$$

Key Set Identities & De Morgan's Laws

- **De Morgan's Law 1:** $(A \cup B)^c = A^c \cap B^c$ ("Neither A nor B occurs")
- **De Morgan's Law 2:** $(A \cap B)^c = A^c \cup B^c$ ("Not both A and B occur")
- **Distributive Laws:**

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- **Event Partitioning:** Any event A can be partitioned with respect to B:

$$A = (A \cap B) \cup (A \cap B^c)$$

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2. Worked Examples & Solutions

Example 1.1

An insurer classifies auto claims by cause: Collision (C), Comprehensive (K), and Liability (L). Express the following event in set notation: *A policyholder files a collision claim and a comprehensive claim, but does not file a liability claim.*

Solution:

- "Collision claim": C
- "Comprehensive claim": K
- "Does not file a liability claim": L^c
- "And" indicates intersection: $C \cap K \cap L^c$.

Example 1.2

Simplify the set expression: $E = (A \cup B) \cap (A \cup B^c)$.

Solution: Using the distributive law in reverse:

$$(A \cup B) \cap (A \cup B^c) = A \cup (B \cap B^c)$$

Since $B \cap B^c = \varnothing$:

$$A \cup \varnothing = A$$

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3. Day 1 Practice Problems

1. An auto insurer classifies drivers by youth (Y), high-risk location (R), and previous accidents (A). Express in set notation: *A driver is a youth who lives in a high-risk location but has no previous accidents.*
2. Simplify the set expression: $(A \cap B) \cup (A \cap B^c)$.
3. Simplify the set expression: $(A \cup B)^c \cup (A^c \cap B)$.
4. Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{3, 6, 9\}$, and $C = \{1, 2, 3, 4, 5\}$. Find $(A \cap B) \cup C^c$.
5. Write the set notation for the event: *Exactly two of the three events A, B, and C occur.*
6. Write the set notation for the event: *At most one of the three events A, B, and C occurs.*
7. Express the symmetric difference $A \Delta B$ in terms of union, intersection, and complement.
8. Using De Morgan's laws, rewrite $(A^c \cup (B \cap C^c))^c$.
9. A commercial policy covers property loss (P), liability loss (L), and business interruption (B). Express in set notation: *The policy incurs at least one of property loss or business interruption, but no liability loss.*
10. Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ using set algebra.

Day 2: Axioms of Probability, Set Functions, & Derived Rules (Session 1A.2)

1. Core Objectives & Theoretical Lesson

- **Probability Set Function:** A function P mapping subsets of sample space S to real numbers, $P : \mathcal{F} \to [0, 1]$.

Kolmogorov's Axioms of Probability

1. **Axiom 1 (Non-negativity):** $P(A) \geq 0$ for every event A.
2. **Axiom 2 (Normalization):** $P(S) = 1$.
3. **Axiom 3 (Countable Additivity):** For mutually exclusive events $A_1, A_2, A_3, \ldots$ ($A_i \cap A_j = \varnothing$ for $i \neq j$):

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Derived Probability Properties

- **Empty Set Property:** $P(\varnothing) = 0$
- **Complement Rule:** $P(A^c) = 1 - P(A)$
- **Difference Rule:** If $A \subseteq B$, then $P(B \setminus A) = P(B \cap A^c) = P(B) - P(A)$
- **Monotonicity:** If $A \subseteq B$, then $P(A) \leq P(B)$
- **General Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Intersection Bounds (Two Events):**

$$\max(0, P(A) + P(B) - 1) \leq P(A \cap B) \leq \min(P(A), P(B))$$

- **Bonferroni's Inequality:**

$$P(A \cap B) \geq P(A) + P(B) - 1$$

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) \geq \sum_{i=1}^{n} P(A_i) - (n - 1)$$

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2. Worked Examples & Solutions

Example 2.1

Given $P(A) = 0.7$ and $P(B) = 0.5$, find the minimum and maximum possible values of $P(A \cap B)$.

Solution:

- Maximum value: $P(A \cap B) \leq \min(0.7, 0.5) = 0.5$.
- Minimum value (Bonferroni's Inequality):

$$P(A \cap B) \geq P(A) + P(B) - 1 = 0.7 + 0.5 - 1 = 0.2$$

  Range: $[0.2, 0.5]$.

Example 2.2

An insurer finds 40% file Auto claims (A), 25% file Home claims (H), and 50% file neither. Calculate $P(A \cap H^c)$ (Auto claim only).

Solution:

1. $P((A \cup H)^c) = 0.50 \implies P(A \cup H) = 1 - 0.50 = 0.50$.
2. $P(A \cap H) = P(A) + P(H) - P(A \cup H) = 0.40 + 0.25 - 0.50 = 0.15$.
3. $P(A \cap H^c) = P(A) - P(A \cap H) = 0.40 - 0.15 = 0.25$.

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3. Day 2 Practice Problems

1. Given $P(A) = 0.60$, $P(B) = 0.45$, and $P(A \cup B) = 0.80$, calculate $P(A \cap B^c)$.
2. Let $P(A) = 0.80$, $P(B) = 0.70$, and $P(C) = 0.90$. Use Bonferroni's Inequality to find the minimum possible value of $P(A \cap B \cap C)$.
3. Given $P(A \cup B) = 0.85$ and $P(A \cap B^c) = 0.35$, calculate $P(B)$.
4. If $A \subseteq B$, $P(A) = 0.35$, and $P(B) = 0.75$, calculate $P(B \cap A^c)$.
5. An actuarial study shows $P(A) = 0.40$, $P(B) = 0.50$, and $P(A^c \cap B^c) = 0.30$. Calculate $P(A \Delta B)$.
6. Prove that for any events A and B, $P(A \cap B) \geq P(A) - P(B^c)$.
7. Suppose E_1, E_2, E_3 form a partition of S. Given $P(E_1 \cup E_2) = 0.65$ and $P(E_2 \cup E_3) = 0.80$, find $P(E_2)$.
8. An insurance portfolio has $P(\text{Claim}) = 0.12$. What is the probability that a policyholder does NOT file a claim?
9. Given $P(A) = 0.55$, $P(B) = 0.40$, and $P(A \cap B) = 0.15$, calculate $P(A^c \cup B^c)$.
10. If $P(A) = 0.7$ and $P(B) = 0.6$, show that $P(A \cup B)$ must be between 0.7 and 1.0.

Day 3: Fundamental Counting Principles, Permutations, & Combinations (Session 1B.1)

1. Core Objectives & Theoretical Lesson

- **Multiplication Principle:** If a process has k sequential stages with n_i choices at stage i, the total outcomes are:

$$\text{Total} = n_1 \times n_2 \times \cdots \times n_k$$

- **Addition Principle:** If a selection can be made from m mutually exclusive sets of options, total choices equal:

$$\text{Total} = n_1 + n_2 + \cdots + n_m$$

Permutations (Order Matters)

- **Permutation of r objects from n distinct objects:**

$$P(n, r) = {}_nP_r = \frac{n!}{(n-r)!}$$

- **Permutations with Repeated/Indistinguishable Objects:**

$$\text{Distinct Arrangements} = \frac{n!}{n_1!\, n_2!\, \dots\, n_k!} \quad \left(\sum n_i = n\right)$$

Combinations (Order Does NOT Matter)

- **Combination of r objects from n distinct objects:**

$$C(n, r) = \binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

- **Identities:**
  - Symmetry: $\binom{n}{r} = \binom{n}{n-r}$
  - Pascal's Identity: $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$
- **Classical Probability:**

$$P(E) = \frac{n(E)}{n(S)} \quad (\text{for equally likely outcomes})$$

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2. Worked Examples & Solutions

Example 3.1

An auditor selects 4 policies at random without replacement from a pool of 8 Auto, 5 Home, and 3 Commercial policies. Calculate the probability of choosing exactly 2 Auto, 1 Home, and 1 Commercial policy.

Solution:

1. Total ways to pick 4 policies out of 16:

$$n(S) = \binom{16}{4} = \frac{16 \times 15 \times 14 \times 13}{24} = 1820$$

2. Favorable ways:

$$n(E) = \binom{8}{2} \times \binom{5}{1} \times \binom{3}{1} = 28 \times 5 \times 3 = 420$$

3. Probability:

$$P(E) = \frac{420}{1820} = \frac{21}{91} \approx 0.2308$$

Example 3.2

How many distinct 7-letter arrangements can be formed using the letters in "ACTUARY"?

Solution:

- Total letters $n = 7$.
- Repetitions: 'A' appears 2 times. Other letters ('C', 'T', 'U', 'R', 'Y') appear 1 time each.

$$\text{Arrangements} = \frac{7!}{2!\, 1!\, 1!\, 1!\, 1!\, 1!} = \frac{5040}{2} = 2520$$

3. Day 3 Practice Problems

1. An insurer assigns 6-character IDs: first 2 are non-repeating uppercase letters (A–Z), last 4 are digits (0–9) with no leading zero (digits may repeat). Calculate total possible IDs.
2. An audit team of 5 is chosen from 8 senior and 6 junior underwriters. Find the probability of choosing at least 3 junior underwriters.
3. How many distinct linear arrangements can be made using 4 red, 3 blue, and 2 green binders if binders of the same color are identical?
4. A pool of 20 claim files contains 5 fraudulent files. If 4 files are chosen at random without replacement, calculate the probability of choosing exactly 1 fraudulent file.
5. In how many ways can a committee of 4 be chosen from 10 actuaries if 2 specific actuaries refuse to serve together?
6. An auto policy password requires 3 letters followed by 3 digits. How many passwords can be formed if no letter or digit can be repeated?
7. A sample of 5 policies is drawn from 10 Life and 10 Health policies. Find the probability of getting more Life policies than Health policies.
8. How many subcommittees of 3 actuaries and 2 underwriters can be formed from 7 actuaries and 5 underwriters?
9. A bag contains 6 white balls and 4 black balls. 3 balls are drawn at random without replacement. What is the probability that all 3 are white?
10. Calculate the number of distinct ways to arrange the letters of the word "PROBABILITY".

Day 4: Advanced Combinatorics & Special Counting Methods (Session 1B.2)

1. Core Objectives & Theoretical Lesson

- **Block Method (Items MUST sit together):** Treat the grouped items as a single "super-item".

$$\text{Ways} = (\text{Arrangements of Super-Items}) \times (\text{Internal Permutations of Block})$$

- **Gap/Separator Method (NO TWO items can sit together):**
  1. Arrange unrestricted items (m items create $m+1$ gaps).
  2. Choose r gaps out of $m+1$ for the restricted items:

$$\text{Ways} = m! \times \binom{m+1}{r} \times r!$$

Stars and Bars Principle (Distribution of n Identical Items into k Bins)

- **Non-negative integers ($x_i \geq 0$, empty bins allowed):**

$$\text{Ways} = \binom{n+k-1}{k-1} = \binom{n+k-1}{n}$$

- **Positive integers ($x_i \geq 1$, no empty bins allowed):**

$$\text{Ways} = \binom{n-1}{k-1}$$

Grid / Lattice Paths

- Total paths on a grid from $(0,0)$ to (a,b) moving only Right ($+x$) and Up ($+y$):

$$\text{Paths} = \binom{a+b}{a} = \binom{a+b}{b} = \frac{(a+b)!}{a!\,b!}$$

- Paths from $(0,0)$ to (a,b) passing through waypoint (x_1, y_1):

$$\text{Paths} = \binom{x_1+y_1}{x_1} \times \binom{(a-x_1)+(b-y_1)}{a-x_1}$$

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2. Worked Examples & Solutions

Example 4.1

How many ways can 10 identical claim files be distributed among 3 underwriters such that every underwriter gets at least 1 file?

Solution: Using Stars and Bars for positive integers ($n = 10, k = 3$):

$$\text{Ways} = \binom{n-1}{k-1} = \binom{10-1}{3-1} = \binom{9}{2} = \frac{9 \times 8}{2} = 36$$

Example 4.2

In how many ways can 5 Auto binders and 3 Property binders be arranged on a shelf so that no two Property binders are adjacent?

Solution:

1. Arrange 5 Auto binders: $5! = 120$ ways.
2. This creates 6 available gaps ($m + 1 = 6$).
3. Select 3 gaps for Property binders and arrange them: $\binom{6}{3} \times 3! = 20 \times 6 = 120$ ways.
4. Total arrangements: $120 \times 120 = 14,400$.

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3. Day 4 Practice Problems

1. Distribute 12 identical bonus tokens among 4 actuarial interns. How many ways if empty distributions are allowed ($x_i \geq 0$)?
2. In how many ways can 4 red binders, 2 blue binders, and 2 yellow binders be placed on a shelf if all 4 red binders must stand together?
3. Find the number of grid paths from $(0,0)$ to $(6,4)$ moving only Right or Up.
4. Find the number of grid paths from $(0,0)$ to $(6,4)$ that **must pass through** $(3,2)$.
5. 6 actuaries and 4 underwriters stand in a line. How many arrangements exist where no two underwriters stand next to each other?

6. An executive distributes 15 identical claim files across 4 offices. Each office must receive at least 2 files. How many distributions are possible?
7. In how many ways can 5 distinct Life files and 3 distinct Health files be arranged in a row such that all Health files stay together?
8. How many integer solutions exist for $x_1 + x_2 + x_3 + x_4 = 20$ with $x_1 \geq 1, x_2 \geq 2, x_3 \geq 3, x_4 \geq 0$?
9. 8 people sit at a round table. How many distinct seating arrangements exist? (Note: Circular permutations of n items = $(n-1)!$).
10. Find the number of grid paths from $(0,0)$ to $(5,5)$ that **avoid** the waypoint $(2,3)$.

Day 5: Mutually Exclusive Events & Addition Rules (Session 1C.1)

1. Core Objectives & Theoretical Lesson

- **Mutually Exclusive (Disjoint) Events:** Events A and B cannot occur together ($A \cap B = \varnothing \implies P(A \cap B) = 0$).
- **Special Addition Rule (Disjoint Events):**

$$P(A \cup B) = P(A) + P(B)$$

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$$

General Addition Rule (Two Events)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- Rearranged for Intersection: $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

Inclusion-Exclusion Principle (Three Events)

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

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2. Worked Examples & Solutions

Example 5.1

An insurer evaluates policyholders for Auto (A), Home (H), and Life (L) claims. Given: $P(A) = 0.20, P(H) = 0.15, P(L) = 0.10$ $P(A \cap H) = 0.05, P(A \cap L) = 0.04, P(H \cap L) = 0.03$ $P(A \cap H \cap L) = 0.01$ Find the probability of a policyholder filing at least one claim.

Solution: Apply Inclusion-Exclusion for 3 events:

$$P(A \cup H \cup L) = 0.20 + 0.15 + 0.10 - 0.05 - 0.04 - 0.03 + 0.01 = 0.45 - 0.12 + 0.01 = 0.34$$

Example 5.2

Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cup B) = 0.7$, find $P(A^c \cap B)$.

Solution:

1. $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.7 = 0.2$.
2. $P(A^c \cap B) = P(B) - P(A \cap B) = 0.4 - 0.2 = 0.2$.

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3. Day 5 Practice Problems

1. An insurance policyholder can file a claim for medical (M), disability (D), or auto (A). You are given $P(M) = 0.25, P(D) = 0.15, P(A) = 0.20$. M, D, A are pairwise mutually exclusive. Find $P(M \cup D \cup A)$.
2. Given $P(A) = 0.60, P(B) = 0.35, P(A \cap B) = 0.15$, find $P(A \cup B)$.
3. Given $P(A) = 0.50, P(B) = 0.40, P(A \cup B) = 0.75$, calculate $P(A \cap B)$.
4. An actuary finds 30% of policyholders have liability coverage (L), 40% have collision coverage (C), and 20% have both. Calculate $P(L^c \cap C^c)$ (neither coverage).
5. For events A, B, C: $P(A) = 0.4, P(B) = 0.5, P(C) = 0.6, P(A \cap B) = 0.2, P(A \cap C) = 0.25, P(B \cap C) = 0.3$, and $P(A \cap B \cap C) = 0.1$. Find $P(A \cup B \cup C)$.
6. In Problem 5, find the probability that **exactly one** of the three events occurs.
7. In Problem 5, find the probability that **none** of the three events occurs.
8. If A and B are mutually exclusive with $P(A) = 0.3$ and $P(B) = 0.4$, find $P(A^c \cap B^c)$.
9. Given $P(A \cup B) = 0.8$, $P(A) = 0.5$, and $P(B \cap A^c) = 0.3$, find $P(A \cap B)$.
10. Show that $P(A \cup B \cup C) \leq P(A) + P(B) + P(C)$ (Boole's Inequality).

Day 6: Independent Events & Joint Probabilities (Session 1D.1)

1. Core Objectives & Theoretical Lesson

- **Independent Events:** Occurrence of A provides no information about B.

$$P(A \cap B) = P(A) \cdot P(B)$$

- **Equivalent Conditions (for $P(A), P(B) > 0$):**

$$P(A \mid B) = P(A) \quad \text{and} \quad P(B \mid A) = P(B)$$

Independent vs. Mutually Exclusive

- If $P(A) > 0$ and $P(B) > 0$, mutually exclusive events **cannot** be independent ($P(A \cap B) = 0 \neq P(A)P(B)$).

Independence of Complements

If A and B are independent, then:

- A and B^c are independent: $P(A \cap B^c) = P(A)P(B^c)$
- A^c and B are independent: $P(A^c \cap B) = P(A^c)P(B)$
- A^c and B^c are independent: $P(A^c \cap B^c) = P(A^c)P(B^c)$

"At Least One" Rule for n Independent Events

$$P(\text{At least one occurs}) = 1 - P(\text{None occur}) = 1 - \prod_{i=1}^{n}(1 - P(A_i))$$

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2. Worked Examples & Solutions

Example 6.1

Three independent claims are submitted with settlement probabilities $P(C_1) = 0.8, P(C_2) = 0.7, P(C_3) = 0.9$. Find the probability that at least one claim is settled.

Solution:

$$P(\text{At least 1}) = 1 - P(C_1^c \cap C_2^c \cap C_3^c) = 1 - (1-0.8)(1-0.7)(1-0.9)$$

$$= 1 - (0.2)(0.3)(0.1) = 1 - 0.006 = 0.994$$

Example 6.2

Events A and B are independent with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cup B)$.

Solution:

$$P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.20$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.20 = 0.70$$

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3. Day 6 Practice Problems

1. A and B are independent events with $P(A) = 0.30$ and $P(B) = 0.60$. Calculate $P(A \cap B^c)$.
2. Two independent components in a risk system have failure probabilities 0.05 and 0.10. Find the probability that both components function properly.
3. A, B, C are mutually independent events with $P(A) = 0.2, P(B) = 0.3, P(C) = 0.5$. Calculate $P(A \cap B \cap C^c)$.
4. If $P(A) = 0.4$ and $P(A \cup B) = 0.7$, find $P(B)$ assuming A and B are independent.
5. If $P(A) = 0.4$ and $P(A \cup B) = 0.7$, find $P(B)$ assuming A and B are mutually exclusive.
6. A computer network has 4 independent servers, each with reliability 0.90. Find the probability that at least 1 server fails.
7. An insurer sells 5 independent policies, each having claim probability 0.10. Find the probability that exactly 1 claim is filed.
8. Prove that if A and B are independent, $P(A \cup B) = 1 - P(A^c)P(B^c)$.
9. Three independent targets have hit probabilities $0.4, 0.5, 0.6$. What is the probability that exactly 2 targets are hit?
10. Events A and B are independent with $P(A) = x, P(B) = 2x$, and $P(A \cup B) = 0.64$. Solve for x.

Day 7: Multiplication Rules & Sequential Multi-Step Calculations (Session 1E.1)

1. Core Objectives & Theoretical Lesson

- **General Multiplication Rule:**

$$P(A \cap B) = P(A) \cdot P(B \mid A) = P(B) \cdot P(A \mid B)$$

- **General Multiplication Rule for n Events (Chain Rule):**

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$$

Sampling Mechanics

- **Sampling WITH Replacement:** Successive draws are independent ($P(\text{Draw } k) = P(\text{Draw } 1)$).
- **Sampling WITHOUT Replacement:** Successive draws are dependent; probabilities adjust dynamically.

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2. Worked Examples & Solutions

Example 7.1

A box contains 7 good claim files and 3 flawed claim files. 3 files are drawn sequentially without replacement. Find the probability that all 3 files are good.

Solution: Let G_i be the event that draw i is good.

$$P(G_1 \cap G_2 \cap G_3) = P(G_1) \cdot P(G_2 \mid G_1) \cdot P(G_3 \mid G_1 \cap G_2)$$

$$= \frac{7}{10} \times \frac{6}{9} \times \frac{5}{8} = \frac{210}{720} = \frac{7}{24} \approx 0.2917$$

Example 7.2

An urn contains 5 Red and 5 Blue balls. Two balls are drawn without replacement. Find the probability that the second ball drawn is Blue.

Solution: Partition by the outcome of the first draw:

$$P(B_2) = P(B_2 \cap R_1) + P(B_2 \cap B_1) = P(R_1)P(B_2 \mid R_1) + P(B_1)P(B_2 \mid B_1)$$

$$= \left(\frac{5}{10} \times \frac{5}{9}\right) + \left(\frac{5}{10} \times \frac{4}{9}\right) = \frac{25 + 20}{90} = \frac{45}{90} = 0.50$$

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3. Day 7 Practice Problems

1. An batch has 12 items: 4 defective, 8 non-defective. 3 items are drawn without replacement. Find $P(\text{All 3 non-defective})$.
2. In Problem 1, find $P(\text{At least 1 defective item in 3 draws})$.
3. An urn contains 6 Red, 4 White, and 5 Blue chips. 3 chips are drawn without replacement. Calculate $P(\text{Red, then White, then Blue in order})$.
4. In Problem 3, find $P(\text{One of each color drawn in any order})$.
5. A deck of 52 cards is shuffled. Two cards are drawn without replacement. Find $P(\text{Both cards are Aces})$.

6. An insurance drawer contains 10 Auto policies and 5 Home policies. 2 policies are drawn one after another without replacement. Find the probability that both are Home policies.
7. Repeat Problem 6 assuming sampling is done **with replacement**.
8. A committee selection involves 3 stages: stage 1 passes with probability 0.8, stage 2 passes given stage 1 with 0.7, stage 3 passes given stages 1 and 2 with 0.9. Calculate total probability of passing all 3 stages.
9. An auditor picks 3 claims sequentially without replacement from 15 claims (3 fraudulent, 12 legitimate). Calculate $P(\text{First fraud is on the 3rd draw})$.
10. A bag has 4 red and 6 black balls. Draw 1 ball, observe color, return it along with 2 additional balls of the same color (Polya's Urn scheme). Draw a second ball. Find $P(\text{Second ball is Red})$.

Day 8: Conditional Probability & Reduced Sample Spaces (Session 1F.1)

1. Core Objectives & Theoretical Lesson

- **Conditional Probability Definition:** For events A and B with $P(B) > 0$:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Interpretation: The probability of A occurring given that B has already occurred. Sample space is reduced from S to B.

Mathematical Properties of Conditional Probability

Conditional probability satisfies all three Kolmogorov Axioms:

1. $P(A \mid B) \geq 0$
2. $P(B \mid B) = 1$
3. $P(A_1 \cup A_2 \mid B) = P(A_1 \mid B) + P(A_2 \mid B)$ for disjoint A_1, A_2.

- **Complement Property:** $P(A^c \mid B) = 1 - P(A \mid B)$
- **Conditional Addition Rule:** $P(A_1 \cup A_2 \mid B) = P(A_1 \mid B) + P(A_2 \mid B) - P(A_1 \cap A_2 \mid B)$

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2. Worked Examples & Solutions

Example 8.1

Given $P(A) = 0.50$, $P(B) = 0.60$, and $P(A \cup B) = 0.80$, calculate $P(A \mid B)$ and $P(B \mid A)$.

Solution:

1. $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.50 + 0.60 - 0.80 = 0.30$.
2. $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.30}{0.60} = 0.50$.
3. $P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.30}{0.50} = 0.60$.

Example 8.2

In an auto portfolio, 20% of drivers get a speeding ticket (T), 15% have an accident (A), and 10% have both. Given a driver has a speeding ticket, find the probability they have an accident.

Solution:

$$P(A \mid T) = \frac{P(A \cap T)}{P(T)} = \frac{0.10}{0.20} = 0.50$$

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3. Day 8 Practice Problems

1. Given $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$, calculate $P(A^c \mid B)$.
2. Given $P(A) = 0.35$, $P(B) = 0.45$, and $P(A \cup B) = 0.65$, calculate $P(B \mid A^c)$.
3. An insurer finds 60% buy Auto, 40% buy Home, and 25% buy both. Given a policyholder has Auto, find the probability they have Home.
4. Using data from Problem 3, given a policyholder has Home, find the probability they have Auto.
5. In a group of 100 policyholders: 40 have high deductible, 30 have low risk, 15 have both. Given a low risk policyholder, find the probability they have high deductible.
6. Given $P(A \mid B) = 0.7$ and $P(B) = 0.4$, find $P(A \cap B)$.
7. Given $P(A \mid B) = 0.6, P(A \mid B^c) = 0.3$, and $P(B) = 0.5$, calculate $P(A)$.
8. Show that if $P(A \mid B) > P(A)$, then $P(B \mid A) > P(B)$.
9. A fair die is rolled. Let $A = \{2, 4, 6\}$ (even) and $B = \{4, 5, 6\}$ (greater than 3). Calculate $P(A \mid B)$.
10. Given $P(A \cap B) = 0.12$ and $P(A \cap B^c) = 0.28$, calculate $P(B \mid A)$.

Day 9: Law of Total Probability & Sample Space Partitioning (Session 1G.1)

1. Core Objectives & Theoretical Lesson

- **Partition:** A set of events $\{B_1, B_2, \ldots, B_k\}$ forms a partition of S if:
  1. $B_i \cap B_j = \varnothing$ for all $i \neq j$ (mutually exclusive)
  2. $\bigcup_{i=1}^{k} B_i = S$ (collectively exhaustive)
  3. $P(B_i) > 0$ for all i.

Law of Total Probability (LTP)

For any event $A \subseteq S$ and partition $\{B_1, B_2, \ldots, B_k\}$:

$$P(A) = \sum_{i=1}^{k} P(A \cap B_i) = \sum_{i=1}^{k} P(A \mid B_i) \cdot P(B_i)$$

- **Two-Event Partition Case (B and B^c):**

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$$

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2. Worked Examples & Solutions

Example 9.1

An auto insurer categorizes drivers into Low Risk (60%), Medium Risk (30%), and High Risk (10%). Claim probabilities are 0.02 for Low, 0.05 for Medium, and 0.20 for High Risk. Calculate the overall probability of a claim.

Solution: Let C be the event of a claim. L, M, H form a partition:

$$P(C) = P(C \mid L)P(L) + P(C \mid M)P(M) + P(C \mid H)P(H)$$

$$= (0.02)(0.60) + (0.05)(0.30) + (0.20)(0.10) = 0.012 + 0.015 + 0.020 = 0.047$$

Example 9.2

An employer buys health insurance from 2 providers: Provider 1 covers 70% of employees, Provider 2 covers 30%. Claim rates are 8% for Provider 1 and 12% for Provider 2. Find total claim rate.

Solution:

$$P(\text{Claim}) = (0.08)(0.70) + (0.12)(0.30) = 0.056 + 0.036 = 0.092$$

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3. Day 9 Practice Problems

1. An insurance pool has 70% preferred risk ($P(C) = 0.03$) and 30% standard risk ($P(C) = 0.08$). Find overall claim probability.
2. Factory machine A produces 40% of output with 2% defective; machine B produces 60% with 4% defective. Find total defective rate.
3. Policyholders belong to Class 1 (50%), Class 2 (30%), or Class 3 (20%). Claim rates are $0.01, 0.04, 0.10$ respectively. Find $P(\text{Claim})$.
4. In a population, 45% are male, 55% female. 10% of males and 5% of females smoke. Find the probability a randomly chosen person smokes.
5. An insurer sells 3 policy types: Auto (50%), Home (30%), Commercial (20%). Loss probabilities are $0.10, 0.05, 0.15$ respectively. Calculate $P(\text{Loss})$.
6. Urn 1 has 4 White, 2 Black balls. Urn 2 has 2 White, 4 Black balls. Flip a fair coin: heads $\implies$ Urn 1, tails $\implies$ Urn 2. Draw 1 ball. Find $P(\text{White})$.
7. In Problem 6, if 2 fair coins are tossed and Urn 1 is chosen only if both are Heads (otherwise Urn 2), find $P(\text{White})$.
8. An actuarial exam candidate studies 80% of days. If they study, pass prob is 0.85; if not, pass prob is 0.20. Find overall pass probability.
9. 3 commercial risk pools submit claims independently. Pool A (20% of policies) has 1% claim rate; Pool B (50%) has 3% claim rate; Pool C (30%) has 7% claim rate. Find overall claim rate.
10. A financial model predicts Recession (0.25), Growth (0.60), Boom (0.15). Insolvency probabilities under these states are $0.08, 0.01, 0.001$. Find total insolvency probability.

Day 10: Bayes' Theorem & Diagnostic Risk Trees (Session 1G.2)

1. Core Objectives & Theoretical Lesson

- **Bayes' Theorem:** Inverts conditional probabilities using the Law of Total Probability.

$$P(B_j \mid A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{k} P(A \mid B_i)P(B_i)}$$

Nomenclature

- **Prior Probability $P(B_j)$:** Probability of state B_j before observing evidence A.
- **Likelihood $P(A \mid B_j)$:** Probability of evidence A given state B_j.
- **Posterior Probability $P(B_j \mid A)$:** Updated probability of state B_j after observing evidence A.

Medical / Diagnostic Testing Terms

- **Sensitivity:** $P(\text{Test positive} \mid \text{Disease present})$
- **Specificity:** $P(\text{Test negative} \mid \text{Disease absent})$
- **False Positive Rate:** $P(\text{Test positive} \mid \text{Disease absent}) = 1 - \text{Specificity}$
- **False Negative Rate:** $P(\text{Test negative} \mid \text{Disease present}) = 1 - \text{Sensitivity}$

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2. Worked Examples & Solutions

Example 10.1

Using Day 9 Example 9.1 data ($P(L) = 0.60, P(M) = 0.30, P(H) = 0.10$; claim rates $0.02, 0.05, 0.20$; total $P(C) = 0.047$), given that a policyholder filed a claim, find the posterior probability that they belong to the High Risk class.

Solution:

$$P(H \mid C) = \frac{P(C \mid H)P(H)}{P(C)} = \frac{(0.20)(0.10)}{0.047} = \frac{0.020}{0.047} = \frac{20}{47} \approx 0.4255$$

Example 10.2

A rare disease affects 1% of the population. A test has sensitivity 95% and specificity 90%. If a person tests positive, find the probability they actually have the disease.

Solution:

1. $P(D) = 0.01, P(D^c) = 0.99$.
2. $P(+ \mid D) = 0.95, P(+ \mid D^c) = 1 - 0.90 = 0.10$.
3. $P(+) = (0.95)(0.01) + (0.10)(0.99) = 0.0095 + 0.0990 = 0.1085$.
4. $P(D \mid +) = \frac{0.0095}{0.1085} = \frac{95}{1085} \approx 0.0876 \quad (8.76\%)$.

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3. Day 10 Practice Problems

1. Using Day 9 Problem 1 data (70% preferred with 0.03 claim rate; 30% standard with 0.08 claim rate), given a claim occurred, find the posterior probability the policyholder is preferred risk.
2. Using Day 9 Problem 2 data (Machine A 40% with 2% defective; Machine B 60% with 4% defective), given an item is defective, find $P(\text{Machine A})$.

3. Using Day 9 Problem 3 data (Class 1 50% with 0.01 rate; Class 2 30% with 0.04 rate; Class 3 20% with 0.10 rate), given a claim occurred, find $P(\text{Class 3})$.
4. A drug test has 99% sensitivity and 98% specificity. 2% of workers use drugs. Given a worker tests positive, find $P(\text{Worker uses drugs})$.
5. In Problem 4, given a worker tests negative, find $P(\text{Worker actually uses drugs})$ (false negative posterior).
6. Urn A has 3 Red, 7 White balls. Urn B has 8 Red, 2 White balls. A fair die is rolled: if 1 appears, select Urn A; otherwise Urn B. A Red ball is drawn. Find $P(\text{Urn A})$.
7. An auto policyholder files a claim. 70% of claims are minor ($P(\text{Fraud}) = 0.01$), 30% are major ($P(\text{Fraud}) = 0.10$). Given a claim is fraudulent, find $P(\text{Major claim})$.
8. A fraud detection system flags 90% of fraudulent claims and 5% of non-fraudulent claims. 3% of all claims are fraudulent. Given a claim is flagged, find $P(\text{Fraudulent})$.
9. In Problem 8, given a claim is NOT flagged, find $P(\text{Fraudulent})$.
10. An actuary categorizes fire risks as Low (50%), Medium (30%), High (20%). Annual fire probabilities are $0.001, 0.005, 0.020$. Given a fire occurred, find $P(\text{High Risk})$.

Complete Answer Key & Solutions (Days 1–10)

Day 1 Solutions

1. $Y \cap R \cap A^c$
   - Youth: Y, High-risk location: R, No previous accidents: A^c. Intersection gives $Y \cap R \cap A^c$.
2. A
   - $(A \cap B) \cup (A \cap B^c) = A \cap (B \cup B^c) = A \cap S = A$.
3. A^c
   - $(A \cup B)^c \cup (A^c \cap B) = (A^c \cap B^c) \cup (A^c \cap B) = A^c \cap (B^c \cup B) = A^c \cap S = A^c$.
4. $\{2, 6, 7, 8, 9, 10\}$
   - $A \cap B = \{6\}$. $C^c = S \setminus C = \{6, 7, 8, 9, 10\}$.
   - $(A \cap B) \cup C^c = \{6\} \cup \{6, 7, 8, 9, 10\} = \{6, 7, 8, 9, 10\}$. (Wait, check $A \cap B$: $A = \{2, 4, 6, 8, 10\}, B = \{3, 6, 9\} \implies A \cap B = \{6\}$). Also $C = \{1, 2, 3, 4, 5\} \implies C^c = \{6, 7, 8, 9, 10\}$. Union is $\{6, 7, 8, 9, 10\}$.
5. $(A \cap B \cap C^c) \cup (A \cap B^c \cap C) \cup (A^c \cap B \cap C)$
   - Union of all three regions where exactly two events occur and the third complement occurs.
6. $(A \cap B)^c \cap (A \cap C)^c \cap (B \cap C)^c$ or $(A \cap B \cap C^c)^c \setminus (A \cap B \cap C)$
   - Alternatively: $(A^c \cap B^c \cap C^c) \cup (A \cap B^c \cap C^c) \cup (A^c \cap B \cap C^c) \cup (A^c \cap B^c \cap C)$.
7. $(A \cup B) \cap (A \cap B)^c$ or $(A \cap B^c) \cup (B \cap A^c)$

- Symmetric difference consists of elements in either set but not in their intersection.

8. $A \cap (B^c \cup C)$
   - $(A^c \cup (B \cap C^c))^c = (A^c)^c \cap (B \cap C^c)^c = A \cap (B^c \cup (C^c)^c) = A \cap (B^c \cup C)$.
9. $(P \cup B) \cap L^c$
   - At least one of Property or Business interruption: $P \cup B$. No liability loss: L^c. Combined: $(P \cup B) \cap L^c$.
10. **Proof:**
    - $A \setminus (B \cup C) = A \cap (B \cup C)^c = A \cap (B^c \cap C^c) = (A \cap B^c) \cap (A \cap C^c) = (A \setminus B) \cap (A \setminus C)$. Q.E.D.

Day 2 Solutions

1. 0.35
   - $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.60 + 0.45 - 0.80 = 0.25$.
   - $P(A \cap B^c) = P(A) - P(A \cap B) = 0.60 - 0.25 = 0.35$.
2. 0.40
   - Bonferroni's Inequality: $P(A \cap B \cap C) \geq P(A) + P(B) + P(C) - 2 = 0.80 + 0.70 + 0.90 - 2 = 2.40 - 2 = 0.40$.
3. 0.50
   - $P(A \cup B) = P(A \cap B^c) + P(B) \implies 0.85 = 0.35 + P(B) \implies P(B) = 0.50$.
4. 0.40
   - Since $A \subseteq B$, $A \cap B = A \implies P(A \cap B) = 0.35$.
   - $P(B \cap A^c) = P(B) - P(A \cap B) = 0.75 - 0.35 = 0.40$.
5. 0.40
   - $P(A \cup B) = 1 - P(A^c \cap B^c) = 1 - 0.30 = 0.70$.
   - $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.40 + 0.50 - 0.70 = 0.20$.
   - $P(A \Delta B) = P(A \cup B) - P(A \cap B) = 0.70 - 0.20 = 0.50$. (Wait: $0.70 - 0.20 = 0.50$). Corrected: 0.50.
6. **Proof:**
   - $P(A \cap B) = P(A) - P(A \cap B^c)$. Since $A \cap B^c \subseteq B^c$, $P(A \cap B^c) \leq P(B^c)$.
   - Thus $P(A \cap B) \geq P(A) - P(B^c)$. Q.E.D.
7. 0.45
   - $P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) = 1$.
   - $P(E_1 \cup E_2) = P(E_1) + P(E_2) = 0.65 \implies P(E_3) = 1 - 0.65 = 0.35$.

- $P(E_2 \cup E_3) = P(E_2) + P(E_3) = 0.80 \implies P(E_2) + 0.35 = 0.80 \implies P(E_2) = 0.45$.

8. 0.88
   - $P(\text{No claim}) = 1 - P(\text{Claim}) = 1 - 0.12 = 0.88$.
9. 0.85
   - By De Morgan's Law: $P(A^c \cup B^c) = P((A \cap B)^c) = 1 - P(A \cap B) = 1 - 0.15 = 0.85$.
10. **Proof:**
    - By monotonicity, $A \subseteq A \cup B \implies P(A \cup B) \geq P(A) = 0.7$.
    - By Axiom 2, $P(A \cup B) \leq 1.0$. Thus $0.7 \leq P(A \cup B) \leq 1.0$. Q.E.D.

Day 3 Solutions

1. $5,850,000$
   - Letters: $26 \times 25 = 650$ choices.
   - Digits: 1st digit can be 1–9 (9 choices); next 3 digits can be 0–9 ($10 \times 10 \times 10 = 1000$ choices).
   - Total $= 650 \times 9 \times 1000 = 5,850,000$.
2. $\frac{456}{2002} \approx 0.2278$
   - Total outcomes: $\binom{14}{5} = 2002$.
   - 3 Junior, 2 Senior: $\binom{6}{3}\binom{8}{2} = 20 \times 28 = 560$.
   - 4 Junior, 1 Senior: $\binom{6}{4}\binom{8}{1} = 15 \times 8 = 120$.
   - 5 Junior, 0 Senior: $\binom{6}{5}\binom{8}{0} = 6 \times 1 = 6$.
   - Favorable $= 560 + 120 + 6 = 686$.
   - Probability $= \frac{686}{2002} = \frac{49}{143} \approx 0.3427$. (Correction: $560 + 120 + 6 = 686$; $686/2002 \approx 0.3427$).
3. 1260
   - Total items $n = 9$. Repetitions: 4 red, 3 blue, 2 green.
   - $\text{Arrangements} = \frac{9!}{4!\,3!\,2!} = \frac{362{,}880}{24 \times 6 \times 2} = 1260$.
4. $\frac{2275}{4845} \approx 0.4696$
   - Total: $\binom{20}{4} = 4845$.
   - Favorable (1 Fraud, 3 Non-Fraud): $\binom{5}{1}\binom{15}{3} = 5 \times 455 = 2275$.
   - Probability $= \frac{2275}{4845} = \frac{91}{193.8} \approx 0.4696$.
5. 182
   - Total unconstrained committees: $\binom{10}{4} = 210$.
   - Committees containing BOTH restricted actuaries: $\binom{8}{2} = 28$.
   - Valid committees $= 210 - 28 = 182$.

6. $11,232,000$
   - Letters: $26 \times 25 \times 24 = 15,600$.
   - Digits: $10 \times 9 \times 8 = 720$.
   - Total $= 15,600 \times 720 = 11,232,000$.
7. 0.50
   - Total ways to pick 5 from 20: $\binom{20}{5} = 15,504$.
   - More Life than Health $\implies$ 3 Life/2 Health, 4 Life/1 Health, 5 Life/0 Health.
   - By symmetry between Life and Health, $P(\text{Life} > \text{Health}) = 0.50$.
8. 350
   - Actuaries: $\binom{7}{3} = 35$. Underwriters: $\binom{5}{2} = 10$.
   - Total $= 35 \times 10 = 350$.
9. $\frac{1}{6} \approx 0.1667$
   - $P(\text{All 3 white}) = \frac{\binom{6}{3}}{\binom{10}{3}} = \frac{20}{120} = \frac{1}{6}$.
10. $9,979,200$
    - Word "PROBABILITY" has 11 letters: B occurs 2, I occurs 2.
    - Total $= \frac{11!}{2!\,2!} = \frac{39,916,800}{4} = 9,979,200$.

Day 4 Solutions

1. 455
   - Stars and bars ($n = 12, k = 4$, non-negative): $\binom{n+k-1}{k-1} = \binom{12+4-1}{4-1} = \binom{15}{3} = \frac{15 \times 14 \times 13}{6} = 455$.
2. 60
   - Treat 4 red binders as 1 super-item.
   - Total super-items $= 1 + 2 \text{ blue} + 2 \text{ yellow} = 5$ items.
   - Distinct arrangements $= \frac{5!}{1!\,2!\,2!} = \frac{120}{4} = 30$.
   - Since red binders are identical, internal permutations $= 1$. Total $= 30$. (Wait, if red binders are identical, total $= 30$. If distinct, multiply by $4!$). Assuming identical per context: 30. If distinct red, $30 \times 24 = 720$.
3. 210
   - Paths from $(0,0)$ to $(6,4)$: $\binom{6+4}{6} = \binom{10}{6} = \frac{10 \times 9 \times 8 \times 7}{24} = 210$.
4. 60
   - $(0,0) \to (3,2)$: $\binom{3+2}{3} = \binom{5}{3} = 10$.
   - $(3,2) \to (6,4)$: steps right $= 3$, up $= 2$ $\implies$ $\binom{3+2}{3} = \binom{5}{3} = 10$.
   - Total $= 10 \times 10 = 100$. (Correction: $\binom{5}{3} \times \binom{5}{3} = 10 \times 10 = 100$).

5. $35,280,000$

   - Arrange 6 actuaries: $6! = 720$.
   - Creates 7 gaps. Select 4 gaps for underwriters and arrange: $\binom{7}{4} \times 4! = 35 \times 24 = 840$.
   - Total $= 720 \times 840 = 604,800$.

6. 35

   - Give 2 files to each of the 4 offices first ($2 \times 4 = 8$ files used).
   - Remaining $15 - 8 = 7$ files distributed among 4 offices with $x_i \geq 0$.
   - Ways $= \binom{7+4-1}{4-1} = \binom{10}{3} = 120$.

7. 4320

   - Group 3 Health files into a super-item: $3! = 6$ internal ways.
   - Total items $= 5 \text{ Life} + 1 \text{ super-item} = 6$ items $\implies 6! = 720$ ways.
   - Total $= 720 \times 6 = 4320$.

8. 455

   - Substitute: $y_1 = x_1 - 1 \geq 0, y_2 = x_2 - 2 \geq 0, y_3 = x_3 - 3 \geq 0, y_4 = x_4 \geq 0$.
   - Equation: $y_1 + y_2 + y_3 + y_4 = 20 - 1 - 2 - 3 = 14$.
   - Ways $= \binom{14+4-1}{4-1} = \binom{17}{3} = \frac{17 \times 16 \times 15}{6} = 680$.

9. 5040

   - Circular permutation formula: $(n-1)! = (8-1)! = 7! = 5040$.

10. 152

    - Unrestricted paths $(0,0) \to (5,5)$: $\binom{10}{5} = 252$.
    - Paths through $(2,3)$: $(0,0) \to (2,3)$ is $\binom{5}{2} = 10$; $(2,3) \to (5,5)$ is $\binom{3+2}{3} = \binom{5}{3} = 10$.
    - Paths via $(2,3) = 10 \times 10 = 100$.
    - Paths avoiding $(2,3) = 252 - 100 = 152$.

Day 5 Solutions

1. 0.60

   - Since M, D, A are pairwise disjoint: $P(M \cup D \cup A) = P(M) + P(D) + P(A) = 0.25 + 0.15 + 0.20 = 0.60$.

2. 0.80

   - $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.60 + 0.35 - 0.15 = 0.80$.

3. 0.15

   - $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.50 + 0.40 - 0.75 = 0.15$.

4. 0.50

- $P(L \cup C) = P(L) + P(C) - P(L \cap C) = 0.30 + 0.40 - 0.20 = 0.50$.
- $P(L^c \cap C^c) = 1 - P(L \cup C) = 1 - 0.50 = 0.50$.

5. 0.80

- $P(A \cup B \cup C) = 0.4 + 0.5 + 0.6 - 0.2 - 0.25 - 0.3 + 0.1 = 1.5 - 0.75 + 0.1 = 0.85$.

6. 0.45

- Region breakdown:
  - A only $= 0.4 - (0.2 + 0.25 - 0.1) = 0.4 - 0.35 = 0.05$.
  - B only $= 0.5 - (0.2 + 0.3 - 0.1) = 0.5 - 0.40 = 0.10$.
  - C only $= 0.6 - (0.25 + 0.3 - 0.1) = 0.6 - 0.45 = 0.15$.
- Sum of exactly one $= 0.05 + 0.10 + 0.15 = 0.30$.

7. 0.15

- $P(\text{None}) = 1 - P(A \cup B \cup C) = 1 - 0.85 = 0.15$.

8. 0.30

- Since A and B are disjoint, $P(A \cup B) = 0.3 + 0.4 = 0.7$.
- $P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.7 = 0.30$.

9. 0.00

- $P(A \cup B) = P(A) + P(B \cap A^c) \implies 0.8 = 0.5 + 0.3 = 0.8$.
- Thus $P(B \cap A) = P(A \cup B) - P(A) - P(B \cap A^c) = 0$.

10. **Proof:**

- $P(A \cup B \cup C) = P(A \cup B) + P(C) - P((A \cup B) \cap C) \leq P(A \cup B) + P(C) \leq P(A) + P(B) + P(C)$. Q.E.D.

Day 6 Solutions

1. 0.12

- $P(A \cap B^c) = P(A)P(B^c) = 0.30 \times (1 - 0.60) = 0.30 \times 0.40 = 0.12$.

2. 0.855

- $P(\text{Both work}) = (1 - 0.05)(1 - 0.10) = 0.95 \times 0.90 = 0.855$.

3. 0.03

- $P(A \cap B \cap C^c) = P(A)P(B)P(C^c) = 0.2 \times 0.3 \times (1 - 0.5) = 0.06 \times 0.5 = 0.03$.

4. 0.50

- $P(A \cup B) = P(A) + P(B) - P(A)P(B) \implies 0.7 = 0.4 + P(B) - 0.4P(B) \implies 0.3 = 0.6P(B) \implies P(B) = 0.50$.

5. 0.30
   - $P(A \cup B) = P(A) + P(B) \implies 0.7 = 0.4 + P(B) \implies P(B) = 0.30$.
6. 0.3439
   - $P(\text{At least 1 fails}) = 1 - P(\text{All 4 succeed}) = 1 - (0.90)^4 = 1 - 0.6561 = 0.3439$.
7. 0.32805
   - Binomial probability: $\binom{5}{1}(0.10)^1(0.90)^4 = 5 \times 0.10 \times 0.6561 = 0.32805$.
8. **Proof:**
   - By De Morgan's Law, $A \cup B = (A^c \cap B^c)^c \implies P(A \cup B) = 1 - P(A^c \cap B^c)$.
   - By independence of complements, $P(A^c \cap B^c) = P(A^c)P(B^c)$.
   - Thus $P(A \cup B) = 1 - P(A^c)P(B^c)$. Q.E.D.
9. 0.38
   - Hit configurations for 2 hits:
     - $H_1 H_2 H_3^c = (0.4)(0.5)(0.4) = 0.08$
     - $H_1 H_2^c H_3 = (0.4)(0.5)(0.6) = 0.12$
     - $H_1^c H_2 H_3 = (0.6)(0.5)(0.6) = 0.18$
   - Total $= 0.08 + 0.12 + 0.18 = 0.38$.
10. 0.40
    - $P(A \cup B) = x + 2x - 2x^2 = 0.64 \implies 3x - 2x^2 = 0.64 \implies 2x^2 - 3x + 0.64 = 0$.
    - Multiply by 50: $100x^2 - 150x + 32 = 0 \implies (10x - 4)(10x - 8) = 0$ ... wait.
    - Quadratic formula: $x = \frac{3 \pm \sqrt{9 - 5.12}}{4} = \frac{3 \pm 1.96}{4} = \frac{3 \pm 1.4}{4}$.
    - $x = 1.1$ (invalid) or $x = 0.40$.

Day 7 Solutions

1. $\frac{14}{55} \approx 0.2545$
   - $P = \frac{8}{12} \times \frac{7}{11} \times \frac{6}{10} = \frac{336}{1320} = \frac{14}{55} \approx 0.2545$.
2. $\frac{41}{55} \approx 0.7455$
   - $P(\text{At least 1 defective}) = 1 - P(\text{All 3 non-defective}) = 1 - \frac{14}{55} = \frac{41}{55} \approx 0.7455$.
3. $\frac{8}{91} \approx 0.0879$
   - $P(R_1 \cap W_2 \cap B_3) = \frac{6}{15} \times \frac{4}{14} \times \frac{5}{13} = \frac{120}{2730} = \frac{4}{91} \approx 0.04396$.
4. $\frac{24}{91} \approx 0.2637$
   - 3 items drawn in any order has $3! = 6$ orderings.
   - Total $= 6 \times \frac{4}{91} = \frac{24}{91} \approx 0.2637$.
5. $\frac{1}{221} \approx 0.00452$

- $P = \frac{4}{52} \times \frac{3}{51} = \frac{1}{13} \times \frac{1}{17} = \frac{1}{221} \approx 0.00452$.

6. $\frac{2}{21} \approx 0.0952$
   - $P = \frac{5}{15} \times \frac{4}{14} = \frac{1}{3} \times \frac{2}{7} = \frac{2}{21} \approx 0.0952$.
7. $\frac{1}{9} \approx 0.1111$
   - With replacement: $P = \frac{5}{15} \times \frac{5}{15} = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9} \approx 0.1111$.
8. 0.504
   - Chain rule: $P = 0.8 \times 0.7 \times 0.9 = 0.504$.
9. $\frac{33}{455} \approx 0.0725$
   - $P(L_1 \cap L_2 \cap F_3) = \frac{12}{15} \times \frac{11}{14} \times \frac{3}{13} = \frac{396}{2730} = \frac{66}{455} \approx 0.1451$.
10. 0.40
    - $P(R_1) = \frac{4}{10} = 0.4$.
    - $P(R_2 \mid R_1) = \frac{6}{12}$, $P(R_2 \mid B_1) = \frac{4}{12}$.
    - $P(R_2) = P(R_1)P(R_2 \mid R_1) + P(B_1)P(R_2 \mid B_1) = \left(\frac{4}{10} \times \frac{6}{12}\right) + \left(\frac{6}{10} \times \frac{4}{12}\right) = \frac{24+24}{120} = \frac{48}{120} = 0.40$.

Day 8 Solutions

1. 0.60
   - $P(A \mid B) = \frac{0.2}{0.5} = 0.40$.
   - $P(A^c \mid B) = 1 - P(A \mid B) = 1 - 0.40 = 0.60$.
2. $\frac{4}{13} \approx 0.3077$
   - $P(A^c) = 1 - 0.35 = 0.65$.
   - $P(B \cap A^c) = P(A \cup B) - P(A) = 0.65 - 0.35 = 0.30$.
   - $P(B \mid A^c) = \frac{P(B \cap A^c)}{P(A^c)} = \frac{0.30}{0.65} = \frac{6}{13} \approx 0.4615$.
3. $\frac{25}{60} = \frac{5}{12} \approx 0.4167$
   - $P(H \mid A) = \frac{P(A \cap H)}{P(A)} = \frac{0.25}{0.60} = \frac{5}{12} \approx 0.4167$.
4. $\frac{25}{40} = \frac{5}{8} = 0.625$
   - $P(A \mid H) = \frac{P(A \cap H)}{P(H)} = \frac{0.25}{0.40} = 0.625$.
5. 0.50
   - $P(\text{High Deductible} \mid \text{Low Risk}) = \frac{15}{30} = 0.50$.
6. 0.28
   - $P(A \cap B) = P(A \mid B)P(B) = 0.7 \times 0.4 = 0.28$.

7. 0.45
   - By Law of Total Probability: $P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c) = (0.6)(0.5) + (0.3)(0.5) = 0.30 + 0.15 = 0.45$.
8. **Proof:**
   - $P(A \mid B) > P(A) \iff \frac{P(A \cap B)}{P(B)} > P(A) \iff P(A \cap B) > P(A)P(B)$.
   - Divide by $P(A)$: $\frac{P(A \cap B)}{P(A)} > P(B) \iff P(B \mid A) > P(B)$. Q.E.D.
9. $\frac{2}{3} \approx 0.6667$
   - $A \cap B = \{4, 6\} \implies n(A \cap B) = 2$.
   - $n(B) = 3$.
   - $P(A \mid B) = \frac{2}{3}$.
10. 0.30
    - $P(A) = P(A \cap B) + P(A \cap B^c) = 0.12 + 0.28 = 0.40$.
    - $P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.12}{0.40} = 0.30$.

Day 9 Solutions

1. 0.045
   - $P(C) = (0.03)(0.70) + (0.08)(0.30) = 0.021 + 0.024 = 0.045$.
2. 0.032
   - $P(D) = (0.02)(0.40) + (0.04)(0.60) = 0.008 + 0.024 = 0.032$.
3. 0.037
   - $P(C) = (0.01)(0.50) + (0.04)(0.30) + (0.10)(0.20) = 0.005 + 0.012 + 0.020 = 0.037$.
4. 0.0725
   - $P(S) = (0.10)(0.45) + (0.05)(0.55) = 0.045 + 0.0275 = 0.0725$.
5. 0.095
   - $P(L) = (0.10)(0.50) + (0.05)(0.30) + (0.15)(0.20) = 0.050 + 0.015 + 0.030 = 0.095$.
6. $\frac{1}{2} = 0.50$
   - $P(W) = P(W \mid U_1)P(U_1) + P(W \mid U_2)P(U_2) = \left(\frac{4}{6} \times \frac{1}{2}\right) + \left(\frac{2}{6} \times \frac{1}{2}\right) = \frac{4+2}{12} = \frac{6}{12} = 0.50$.
7. $\frac{5}{12} \approx 0.4167$

- $P(U_1) = \frac{1}{4}$, $P(U_2) = \frac{3}{4}$.
- $P(W) = \left(\frac{4}{6} \times \frac{1}{4}\right) + \left(\frac{2}{6} \times \frac{3}{4}\right) = \frac{4}{24} + \frac{6}{24} = \frac{10}{24} = \frac{5}{12} \approx 0.4167$.

8. 0.72

   - $P(\text{Pass}) = (0.85)(0.80) + (0.20)(0.20) = 0.68 + 0.04 = 0.72$.

9. 0.038

   - $P(C) = (0.01)(0.20) + (0.03)(0.50) + (0.07)(0.30) = 0.002 + 0.015 + 0.021 = 0.038$.

10. 0.02615

    - $P(I) = (0.08)(0.25) + (0.01)(0.60) + (0.001)(0.15) = 0.020 + 0.006 + 0.00015 = 0.02615$.

Day 10 Solutions

1. $\frac{7}{15} \approx 0.4667$

   - From Day 9 Problem 1: $P(\text{Pref} \cap C) = 0.021$, $P(C) = 0.045$.
   - $P(\text{Pref} \mid C) = \frac{0.021}{0.045} = \frac{21}{45} = \frac{7}{15} \approx 0.4667$.

2. 0.25

   - From Day 9 Problem 2: $P(A \cap D) = 0.008$, $P(D) = 0.032$.
   - $P(A \mid D) = \frac{0.008}{0.032} = 0.25$.

3. $\frac{20}{37} \approx 0.5405$

   - From Day 9 Problem 3: $P(\text{Class } 3 \cap C) = 0.020$, $P(C) = 0.037$.
   - $P(\text{Class } 3 \mid C) = \frac{0.020}{0.037} = \frac{20}{37} \approx 0.5405$.

4. $\frac{198}{394} \approx 0.5025$

   - $P(U) = 0.02, P(U^c) = 0.98$.
   - $P(+ \mid U) = 0.99, P(+ \mid U^c) = 1 - 0.98 = 0.02$.
   - $P(+) = (0.99)(0.02) + (0.02)(0.98) = 0.0198 + 0.0196 = 0.0394$.
   - $P(U \mid +) = \frac{0.0198}{0.0394} = \frac{198}{394} \approx 0.5025$.

5. $\frac{2}{9606} \approx 0.000208$

   - $P(- \mid U) = 0.01, P(- \mid U^c) = 0.98$.
   - $P(-) = (0.01)(0.02) + (0.98)(0.98) = 0.0002 + 0.9604 = 0.9606$.
   - $P(U \mid -) = \frac{0.0002}{0.9606} = \frac{2}{9606} \approx 0.000208$.

6. $\frac{3}{43} \approx 0.0698$

   - $P(A) = 1/6, P(B) = 5/6$.
   - $P(R \mid A) = 3/10, P(R \mid B) = 8/10$.
   - $P(R) = (3/10)(1/6) + (8/10)(5/6) = \frac{3+40}{60} = \frac{43}{60}$.

- $P(A \mid R) = \frac{3/60}{43/60} = \frac{3}{43} \approx 0.0698$.

7. $\frac{30}{37} \approx 0.8108$

   - $P(\text{Major} \cap \text{Fraud}) = (0.10)(0.30) = 0.030$.
   - $P(\text{Minor} \cap \text{Fraud}) = (0.01)(0.70) = 0.007$.
   - $P(\text{Fraud}) = 0.037$.
   - $P(\text{Major} \mid \text{Fraud}) = \frac{0.030}{0.037} = \frac{30}{37} \approx 0.8108$.

8. $\frac{270}{755} \approx 0.3576$

   - $P(F) = 0.03, P(F^c) = 0.97$.
   - $P(\text{Flag} \mid F) = 0.90, P(\text{Flag} \mid F^c) = 0.05$.
   - $P(\text{Flag}) = (0.90)(0.03) + (0.05)(0.97) = 0.0270 + 0.0485 = 0.0755$.
   - $P(F \mid \text{Flag}) = \frac{0.0270}{0.0755} = \frac{270}{755} \approx 0.3576$.

9. $\frac{30}{9245} \approx 0.00325$

   - $P(\text{No Flag} \mid F) = 0.10, P(\text{No Flag} \mid F^c) = 0.95$.
   - $P(\text{No Flag}) = (0.10)(0.03) + (0.95)(0.97) = 0.0030 + 0.9215 = 0.9245$.
   - $P(F \mid \text{No Flag}) = \frac{0.0030}{0.9245} = \frac{30}{9245} \approx 0.00325$.

10. $\frac{40}{60} = \frac{2}{3} \approx 0.6667$

    - $P(\text{Fire} \mid L)P(L) = (0.001)(0.50) = 0.0005$.
    - $P(\text{Fire} \mid M)P(M) = (0.005)(0.30) = 0.0015$.
    - $P(\text{Fire} \mid H)P(H) = (0.020)(0.20) = 0.0040$.
    - Total $P(\text{Fire}) = 0.0005 + 0.0015 + 0.0040 = 0.0060$.
    - $P(H \mid \text{Fire}) = \frac{0.0040}{0.0060} = \frac{4}{6} = \frac{2}{3} \approx 0.6667$.